

PRACTICAL ASSIGNMENT [100 MARKS]

Design and implement a DevOps-Enabled Qualification Verification System Using Git and CI/CD Practices

Background

Educational institutions, professional bodies, and employers increasingly require reliable mechanisms to verify academic and professional qualifications. Software engineers must develop systems that are secure, scalable, maintainable and supported by modern DevOps practices.

This assignment requires students to design, develop, test, and deploy a collaborative software solution that enables qualification verification while demonstrating industry-standard DevOps and software engineering principles.

Learning Outcomes

Upon successful completion of this assignment, students will be able to:

1. Apply collaborative software engineering practices using Git.
2. Design and implement a software system that supports qualification verification.
3. Apply DevOps principles through Continuous Integration and Continuous Delivery pipelines.
4. Develop automated tests to verify software requirements.
5. Produce evidence-based evaluations of software quality and team collaboration.
6. Critically justify architectural and technological decisions.

Assignment Requirements

Working in teams of **3-5 members**, develop a qualification verification system using either **Java** or **Python**.

The system should allow authorised users to:

- Register qualifications or certifications.
- Search and retrieve qualification records.
- Verify the authenticity of qualifications.
- Maintain an auditable history of verification activities.

Technical Requirements

1. Version Control and Collaboration

The project must demonstrate:

- Git-based version control
- Branching strategy
- Pull requests and code reviews
- Issue tracking
- Meaningful commit history
- Merge conflict management

2. DevOps Implementation

Implement:

- Continuous Integration pipeline
- Automated build process
- Automated testing
- Continuous Delivery pipeline
- Automated quality checks

Examples:

- GitHub Actions
- GitLab CI/CD
- Jenkins
- Azure DevOps

3. Software Quality Assurance

The system must include:

- Unit tests
- Integration tests
- Test coverage reporting
- Static code analysis
- Coding standards compliance

4. Automated Verification of Requirements

Students must demonstrate how software requirements are automatically verified through:

- Test cases
- Validation rules
- CI/CD quality gates
- Verification reports

Evidence from pipeline executions must be included.

5. Deployment

Deploy the system to an accessible environment using one of the following:

- Docker
- Cloud Platform (AWS, Azure, GCP)

Optional Advanced Features (Bonus)

Up to **10 bonus marks** may be awarded for:

- Agentic AI integration
- Blockchain-based credential verification
- Advanced security mechanisms
- Real-time monitoring dashboards
- Infrastructure-as-Code implementation

Deliverables

1. Git Repository (20%)

Repository must include:

- Source code
- Branch history
- Pull requests
- Documentation
- CI/CD configuration

2. Working Software System (25%)

Assessment criteria:

- Functional completeness
- Usability
- Reliability
- Maintainability

3. Technical Report (25%)

Approximately **3,000-4,000 words**, covering:

- Problem analysis
- Requirements
- System architecture
- Design decisions
- DevOps workflow
- Testing strategy
- Verification strategy
- Critical evaluation

4. Demonstration and Viva (15%)

10–15-minute video presentation demonstrating:

- System functionality
- Git workflow
- CI/CD pipeline
- Automated testing
- Qualification verification process

5. Individual Contribution Report (15%)

Each student must provide:

- Contributions made
- Git activity evidence
- Reflection on lessons learnt; Challenges encountered